

Kim Knudsen, Technical University of Denmark

Introduction to Inverse Problems and Imaging (02624)

Week 1

Welcome - course guide

DTU Compute 2025

Kim Knudsen

02624 Inverse Problems and Imaging

Course guide

Welcome to the course "Inverse Problems and Imaging" at DTU!

Content: In this course we consider the fascinating topics of inverse problems and imaging. This is a broad topic that concerns the determination and computation of physical phenomena from indirect observations, and the reconstruction of images in various contexts from available data. Examples include medical imaging, seismology, remote sensing, heat control and signal processing.

In the course we will both develop mathematical theory, consider several model problems, and do numerical computations. "Inverse Problems and Imaging" is a rapidly expanding and multidisciplinary scientific field bridging applications, mathematical analysis and scientific computing.

Prerequisites: The course is an MSc course and requires basic mathematical maturity and training (e.g. 01035) and some experience with numerical computations (e.g. 02601). A challenge in the course is that most theory and applications are best described in an infinite dimensional setting by compact operators in Hilbert spaces and using partial differential equations, and hence some familiarity with functional analysis and PDE's is a plus. The same goes for mathematical and numerical optimization. Specific knowledge is not required and therefore we will bridge possible gaps as the course unfolds.

Program: The course is scheduled for Wednesdays 13-17 and takes place in 358/045. I will generally use a shifted method for teaching:

11:00-13:30: Recap of last week's lecture and new perspectives
13:30-15:30: Problem solving session
15:45-17:00: Lecture on new material

Every week we prepare a specific reading list and a list of exercises. We will try to have the plan ready on Friday afternoon for the coming week. Some weeks have a different program where we work on HW problems or have invited guests giving research-oriented lectures.

Preparation: We strongly recommend that you **review** the program for the course day before arriving. This includes getting an overview of the text material provided. After the lecture (before next week) you will have time to complete exercises and read the material in detail with focus on aspects that you find particularly interesting or aspects where you need to dive deeper to get the full picture.

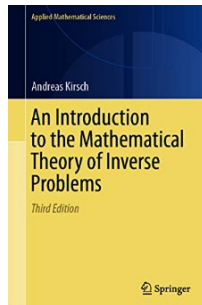
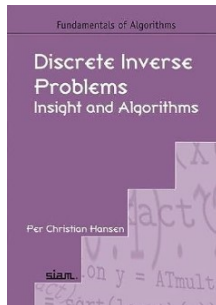
Material: No single book covers the material that we consider. We will provide and use excerpts from two books, see below. Unfortunately, the notation will change slightly from source to source.

[Kirsch]: Andreas Kirsch, An introduction to the mathematical theory of inverse problems, 3rd ed., Springer 2021. Available online via <https://books.dtu.dk/files/02624/02624-13021>.

Kirsch' book discusses inverse problems in the framework of functional analysis.

[FOI]: Per Christian Hansen, Discrete Inverse Problems, Insight and Algorithms, SIAM 2010. An electronic draft will be provided via Learn; do not circulate.

Hansen treats inverse problems with an eye to discretization and computations. And brings great intuition.

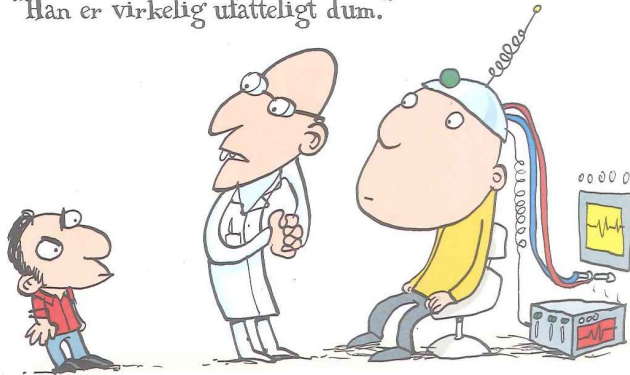


Practical information

- Program
 - 13:00–13:30 Recap of last weeks material (slides)
 - 13:40–15:40 Problem solving session (pen&paper + computer)
 - 15:50–17:00 New material (BB)
- Assessment: Mini-projects + oral exam; pass/no pass
- Generative AI

A collage of mathematical symbols and formulas, including $f(x+\Delta x) = \sum_{i=0}^{\infty} \frac{(\Delta x)^i}{i!} f^{(i)}(x)$, $\int_a^b \epsilon$, Θ , $\sqrt{17}$, Ω , $\int \delta e^{i\pi} = -1$, ∞ , χ , Σ , and a large exclamation mark.

“Han er virkelig ufatteligt dum.”



Jakob Martin Strid: Den dumme mand

What is “Inverse Problems”?

Inverse problems are a large and emerging field in applied science

- Interdisciplinary and applications
- Hard mathematical problems
- Computational aspects: how to compute appropriate solution

From the books...

Keller (1976) *We call **two problems inverse of one another** if the formulation of one problem involves all or part of the solution of the other*

Engl, Hanke and Neubauer (1996) *Inverse problems are concerned with determining **causes** for a desired or observed **effect***

Isakov (2006) *An Inverse Problem assumes a direct problem that is a **well-posed problem** of mathematical physics*

Inverse problems concerns the determination of a phenomenon from indirect measurements and the forward model

What is “Imaging”?

Computing images from data

What is an image?

Dimension

Analogue vs. digital



Example 1: Thermometer

From mercury column to temperature.

Forward mapping F takes temperature T to mercury column height $h = \sigma T$

$$F : T \mapsto \sigma T$$

Forward problem: given T compute $F(T) = \sigma T$.

Inverse problem: Given observation $h = \sigma T$ find T .

Example 2: Rootfinding

Direct problem: given N numbers $x_1, x_2, \dots, x_N$, find a polynomial that has these as roots.

$$P(x) = (x - x_1)(x - x_2) \cdots (x - x_N).$$

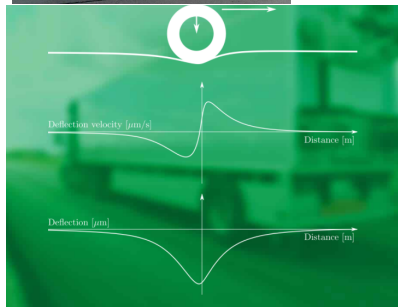
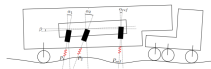
Inverse problem: given a polynomial of degree N

$$P_N(x) = x^N + a_{N-1}x^{N-1} + \cdots + a_1x + a_0,$$

find the N roots of it (counting multiplicity).

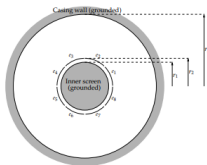
Example 3: Greenwood Engineering TSD

From measurements of road deflection bassin, infer the condition of the lower layers of the road (courtesy of Louis Pedersen)



Example 4: Welltec ECT

From electrostatic electrode measurements, determine the flow in an oil pipe (courtesy of Jimmy Kjærsgård-Rasmussen)



Linear Backprojection



Image 3



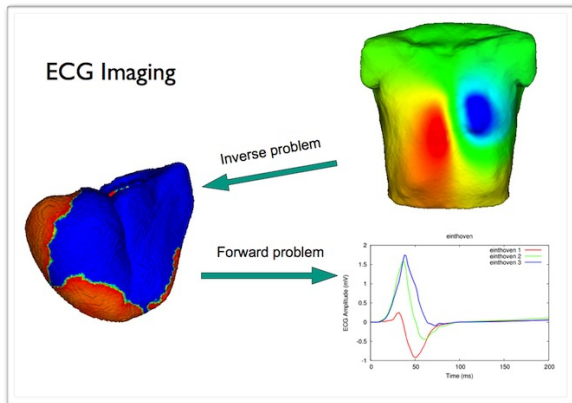
Linear Backprojection



Image 10



Example 5: The Inverse Problems in ECG



<https://www.ibt.kit.edu/>

Example 6: Electrical Impedance Tomography



We will in the course focus on

- Mathematical and computational modelling
- Mathematical analysis
- Hands on computational experiments

Break out(5 min)

Groups of 2-3 students:

Find 1-2 examples of inverse problems from your daily life.

Mathematical setup

Spaces

X : Phenomena

Y : Measurements

and a model

$$F : X \rightarrow Y$$

that maps desired phenomenon to the measurement.

Forward problem: given $x \in X$ find $y = F(x)$.

Inverse problem: given observation of y , find x such that

$$F(x) = y.$$

Model problem 0: Differentiation

Forward problem: given a function x find the primitive y with $y(0) = 0$ such that $y' = x$.

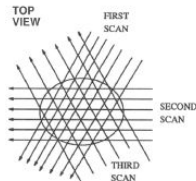
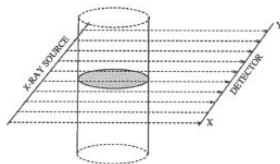
Forward problem:

$$F : x \mapsto y(t) = \int_0^t x(s) ds$$

Inverse problem: given a function y find the derivative $y' = x$.

Model problem 1: Computerized Tomography

Imaging of a body from x-ray measurements.



Parallel beam computerized tomography

(Loading sinogram.mp4)

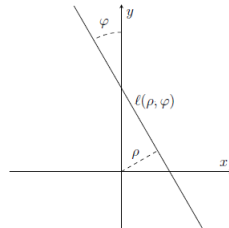
X-rays are send through the object at different angles; the attenuation along a ray is observed.

(Courtesy of Prof. Samuli Siltanen from University of Helsinki - check YouTube!)

Model problem 1

Line integrals of attenuation $a(z_1, z_2)$

$$p(\varphi, \rho) = \int_{\ell(\rho, \varphi)} a(z_1, z_2) ds$$



Rotational symmetry

When $a(z_1, z_2) = a(\sqrt{z_1^2 + z_2^2}) = a(r)$, then $p(\varphi, \rho) = p(\rho)$ is independent of φ and we get

$$a(r) \mapsto p(\rho)$$

Changing variables

$$y(t) = p(\sqrt{1-t}) \quad x(s) = a(\sqrt{1-s})$$

gives the Abel integral operator equation

$$Kx(t) = \int_0^t \frac{1}{\sqrt{t-s}} x(s) ds = y(t).$$

Model problem 2: Backwards Heat Equation

In a uniform metal rod/wire: From temperature distribution at terminal time $t = T > 0$ the temperature distribution at preceeding time $t = 0$.

Model

$$\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} = 0, \quad 0 < x < \pi, \quad t > 0$$

Boundary conditions: $u(0, t) = u(\pi, t) = 0$

Initial condition: $u(x, 0) = f(x)$.

Forward problem: from initial temperature $f(x)$ compute

$$g(x) = u(x, T).$$

Inverse problem: from terminal distribution g find f .

Analysis of the forward problem I

Take the sine-basis $\{\sqrt{\frac{2}{\pi}} \sin(nx)\}$ in $L^2(0, \pi)$ and expand

$$f(x) = \sqrt{\frac{2}{\pi}} \sum_{n=1}^{\infty} \hat{f}(n) \sin(nx),$$

where $\hat{f}(n)$, $n = 1, 2, \dots$ are the Fourier coefficients of $f(x)$:

$$\hat{f}(n) = \sqrt{\frac{2}{\pi}} \int_0^{\pi} f(y) \sin(ny) dy.$$

Using separation of variables the unique solution to the forward problem is

$$g(x) = u(x, T) = \sqrt{\frac{2}{\pi}} \sum_{n=1}^{\infty} \hat{f}(n) e^{-Tn^2} \sin(nx).$$

Analysis of the forward problem II

Interchanging sum and integration gives (with variables $s = y, t = x$)

$$g(s) = Kf(s) = \int_0^\pi k(s, t)f(t) \, dy$$

with

$$k(s, t) = \frac{2}{\pi} \sum_{n=1}^{\infty} e^{-Tn^2} \sin(nx) \sin(ny).$$

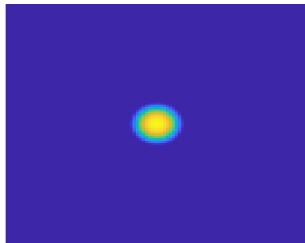
Model problem 3: Image deblurring

Deblurring of images due to lens out of focus, imperfections...

 f  $g = Kf$

$$Kf(y) = \int \phi(x - y)f(y)dy$$

Convolution $Kf(x) = \int \phi(x - y)f(y)dy$



Model problem 3: Phillips 1D

Define

$$\phi(z) = \begin{cases} 1 + \cos(\pi z/3), & |z| < 3, \\ 0, & |z| \geq 3. \end{cases}$$

The model problem is then defined via the convolution operator K

$$Kf(s) = \int_{-6}^6 \phi(s-t)f(t)dt.$$

Inverse and Ill-posed problems

Given model F and measurement/data g , solve

$$F(f) = g.$$

Three questions for well-posedness:

Existence : For what g 's can we solve the equation?

Uniqueness : Does $F(f_1) = F(f_2)$ imply $f_1 = f_2$? (injectivity)

Stability: Can we estimate $\|f - f_\varepsilon\| \leq C\|g - g_\varepsilon\|$?

Hadamard (1923): If one condition fails, the problem is called *ill-posed*.

Interesting inverse problems are very often ill-posed

Linear inverse problems in Hilbert spaces

Let X, Y be Hilbert spaces (think $\mathbb{R}^n$ or L^2). Suppose $K \in B(X, Y)$ and consider for $f \in X$ and $g \in Y$

$$Kf = g.$$

- Forward problem: given K and f , compute g .
- Inverse problem: given K and g , find f .

The forward problem is in this formulation born to be well-posed.

Compact operators

Common framework for many inverse problems

$$Kf(s) = \int_a^b k(s, t)f(t) dt.$$

Such a K is *compact* in $L^2(a, b)$, when $\int_a^b \int_a^b |k(s, t)|^2 ds dt < \infty$.

Operator equation $Kf = g$ is well approximated by a discrete inverse problem

$$Ax = y, \quad A \in \mathbb{R}^{m \times n}, \quad y \in \mathbb{R}^m.$$

Challenge: When $\text{Ran}(K)$ has infinite dimension, K^{-1} cannot be bounded; the inverse problem is born to be ill-posed.